

You can use the following table as general guidance for the average percentage of frequency of letters during common English usage:

e: 12.7	t: 9.1	a: 8.2	o: 7.5	i: 7.0	n: 6.9	s: 6.3	h: 6.1	r: 6.0
d: 4.2	l: 4.0	c: 2.8	u: 2.8	m: 2.4	w: 2.4	f: 2.2	g: 2.0	y: 2.0
p: 1.9	b: 1.5	v: 1.0	k: 0.8	j: 0.2	x: 0.2	q: 0.1	z: 0.1	

Transposition systems

Transposition systems are fundamentally different from substitution systems. In substitution systems, plaintext values are replaced with other values. In transposition systems, plaintext values are rearranged without otherwise changing them. This changes the approach you must adopt when attempting to decrypt these codes.

If you believe that the code you're trying to decrypt uses a transposition system for encryption, arrange the letters in the form of a grid. By trying different variations of number of rows against number of columns, as well as reading off the letters in varying fashion (horizontally or vertically), one will eventually be able to decode the message into plaintext. However, this method can be quite laborious and painstaking, depending on the size of the message to be decoded and the number of letters/variables involved.

Catch phrases

Frequently, many messages are passed off in code as common language. A notable example of this was the British Broadcasting Corporation's overseas service's use of "personal messages" as part of its regular broadcast schedule. The seemingly nonsensical messages read out by announcers were actually one time codes intended for Special Operations Executive agents operating behind enemy lines. Using such means, the French Resistance was instructed to start sabotaging rail and other transport links the night before the D-Day Invasion of Normandy.

The problem with cryptanalysis of these codes is that they are heavily reliant on the cryptanalyst's knowledge. If the cryptanalyst is unaware of the words to watch out for, the 'key' that indicates it is a coded message, he/she will have to attempt catching the code simply by intuition. Again, the lack of a key will force the cryptanalyst to use brute-force methods to try and decrypt the code, rendering it an extremely time-consuming and prohibitive process. Common catch phrases are: salutations, "please find enclosed", etc which are typical in official communication.